

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460929.6723 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.024
Transit depth (Rp/Rs)^2	2.32 +/- 0.73 [%]
Orbital Inclination (inc)	89.47 +/- 0.85
Airmass coefficient 1 (a1)	0.7172 +/- 0.0099
Airmass coefficient 2 (a2)	0.262 +/- 0.076
Scatter in the residuals of the lightcurve fit is	1.39 %
Best Comparison Star	#1 - [553, 231]
Optimal Aperture	4.45
Optimal Annulus	9.22
Transit Duration (day)	0.0872 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.024 vs 0.1714 (deviation 0.81σ)
Duration fit vs published	0.0872 d vs 0.0892 d (deviation 0.63σ)
Mid-time O–C	-3.7 min
Depth SNR	6.3
Residual scatter	1.39 %

Light curve

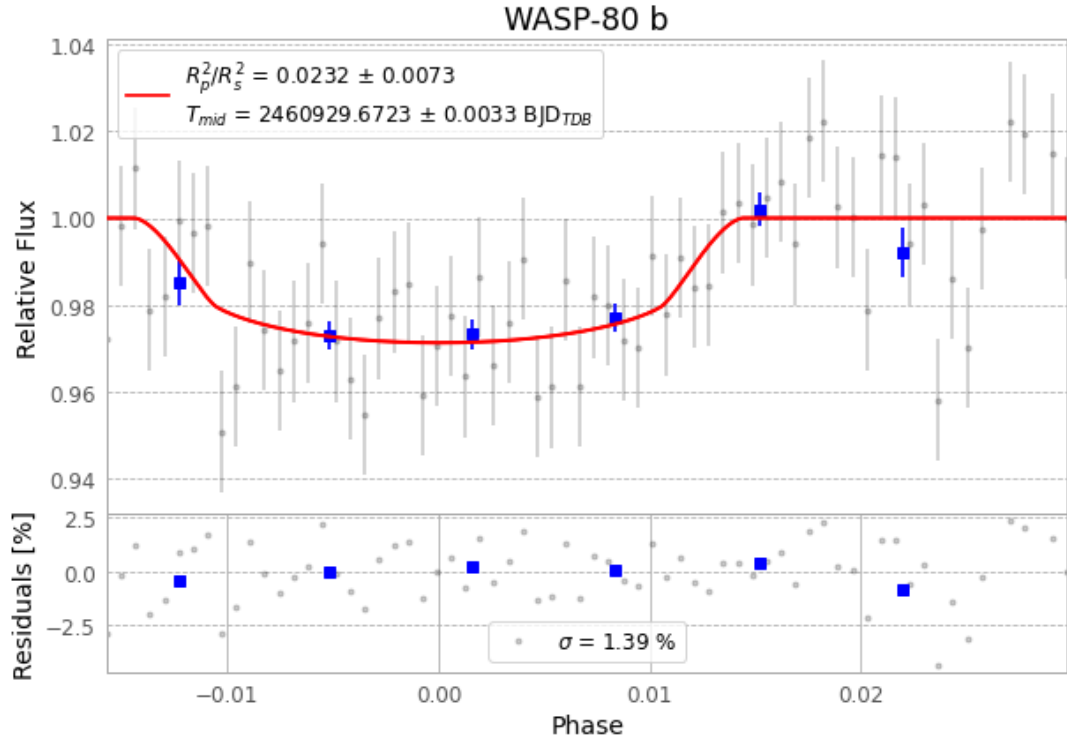


Figure 1 — FinalLightCurve_WASP-80 b_2025-09-10.png (SHA-256 82cb99c11c78bb82...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [321, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7002d68140eb6f80...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), WASP-80250911025115.FITS → WASP-80250911061215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-250911.log (bcc747827239...), FinalLightCurve_WASP-80 b_2025-09-10.png (82cb99c11c78...), FinalLightCurve_WASP-80 b_2025-09-10.csv (e28b5c35cfc4...)

Compute resources

Wall time 153.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 01:04Z.